

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2025 4WM1H Worked Solutions

Nov2025 | 4WM1H | Unit 1

37 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 2**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 1 The diagram shows a semicircle.

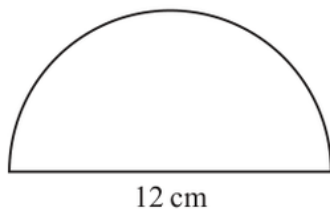


Diagram **NOT**
accurately drawn

The semicircle has diameter 12 cm

Work out the perimeter of the semicircle.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 1 is 2 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 2**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Radius = 6 cm.

$$\text{perimeter} = \pi r + 12 = 6\pi + 12 \approx 30.8496$$

FINAL ANSWER

30.8 cm to 3 significant figures.

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

Round each number to one significant figure:

$$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$$

$$\begin{aligned} \frac{2.11^2 \times 58.9}{\sqrt{8.859}} &\approx \frac{2^2 \times 60}{\sqrt{9}} \\ &= \frac{4 \times 60}{3} = 80 \end{aligned}$$

FINAL ANSWER

80

PAST PAPER QUESTION**Q#: 3****Marks: 3**TOPIC: **Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

3 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 3 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 3****Marks: 3**TOPIC: **Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Algebraic proof. The tag is acceptable because the question asks to show the exact result.

Solution

$$\frac{3}{7} + \frac{1}{3}$$

Use denominator 21:

$$\frac{3}{7} = \frac{9}{21}, \quad \frac{1}{3} = \frac{7}{21}$$

$$\frac{9}{21} + \frac{7}{21} = \frac{16}{21}$$

FINAL ANSWER

$$\frac{16}{21}$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

4 $\mathcal{E} = \{8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18\}$

$A = \{\text{odd numbers}\}$

$B = \{\text{multiples of 3}\}$

$C = \{\text{prime numbers}\}$

(a) List the members of the set $A \cup C$

.....
(1)

(b) List the members of the set $(A \cap B)'$

.....
(1)

(c) Is it true that $15 \in (B \cap C)$?

Tick (✓) the correct box.

Yes

☐

No

☐

Give a reason for your answer.

.....
(1)

D is a set such that $D \cup C = A$

D has 2 members.

(d) List the members of D

.....
(1)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 4****TOPIC: Set Notation & Venn Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$A = \{9, 11, 13, 15, 17\}, \quad B = \{9, 12, 15, 18\}, \quad C = \{11, 13, 17\}$$

$$A \cup C = \{9, 11, 13, 15, 17\}$$

$$A \cap B' = \{11, 13, 17\}$$

$15 \notin B \cap C$ because 15 is not prime.

Since $D \cup C = A$ and D has two members,

$$D = \{9, 15\}$$

FINAL ANSWER

$\{9, 11, 13, 15, 17\}$, $\{11, 13, 17\}$, No, $D = \{9, 15\}$.

PAST PAPER QUESTION**Q#: 5** **Marks: 5****TOPIC: Probability Toolkit**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 5 All students at a school are asked which sport they prefer from football, tennis, rugby and cricket.

The table gives information about the probabilities that, when a student is chosen at random, the student prefers football or tennis or rugby or cricket.

Favourite sport	football	tennis	rugby	cricket
Probability	$2x$	0.2	x	0.35

One of the students at the school is chosen at random.

- (a) Work out the probability that this student prefers tennis or cricket.

.....
(1)

There are 700 students at the school.

- (b) Work out how many students prefer football.

.....
(4)

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 5**TOPIC: **Probability Toolkit**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$P(\text{tennis or cricket}) = 0.2 + 0.35 = 0.55$$

Also,

$$2x + 0.2 + x + 0.35 = 1$$

$$3x = 0.45$$

$$x = 0.15$$

So

$$P(\text{football}) = 2x = 0.30$$

$$700 \times 0.30 = 210$$

FINAL ANSWER

0.55, 210 students.

PAST PAPER QUESTION**Q#: 6****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

6 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 6 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 6****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

The weight 475 g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

The height 120 cm correct to the nearest 10 cm has upper bound

$$120 + 5 = 125$$

FINAL ANSWER

(a) 474.5 g, (b) 125 cm

PAST PAPER QUESTION**Q#: 7****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

7 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n $n = \dots\dots\dots$

(Total for Question 7 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 7****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic roots and indices. The tag is correct.**Solution**

$$\frac{8^{-2} \times 8^9}{8^{10}} = 8^{-2+9-10} = 8^{-3}$$

FINAL ANSWER

$$n = -3$$

PAST PAPER QUESTION**Q#: 8****Marks: 9**TOPIC: **Factorising**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

8 (a) Expand $5w(3w - 4w^2)$
(2)(b) Simplify fully $(4a^3c^5)^3$
(2)(c) Factorise fully $14m^3p^2 + 21m^2p^4$
(2)(d) (i) Factorise $x^2 - 10x + 24$
(2)(ii) Hence solve $x^2 - 10x + 24 = 0$
(1)

(Total for Question 8 is 9 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 9****TOPIC: Factorising**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

(a)

$$5w(3w - 4w^2) = 15w^2 - 20w^3$$

(b)

$$(4a^3c^5)^3 = 64a^9c^{15}$$

(c)

$$14m^3p^2 + 21m^2p^4 = 7m^2p^2(2m + 3p^2)$$

(d)

$$x^2 - 10x + 24 = (x - 4)(x - 6)$$

So

$$x = 4 \quad \text{or} \quad x = 6$$

FINAL ANSWER
 $15w^2 - 20w^3, 64a^9c^{15}, 7m^2p^2(2m + 3p^2), (x - 4)(x - 6), x = 4, 6.$

PAST PAPER QUESTION**Q#: 9** **Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 9 The diagram shows a sketch of a right-angled triangle, ABC

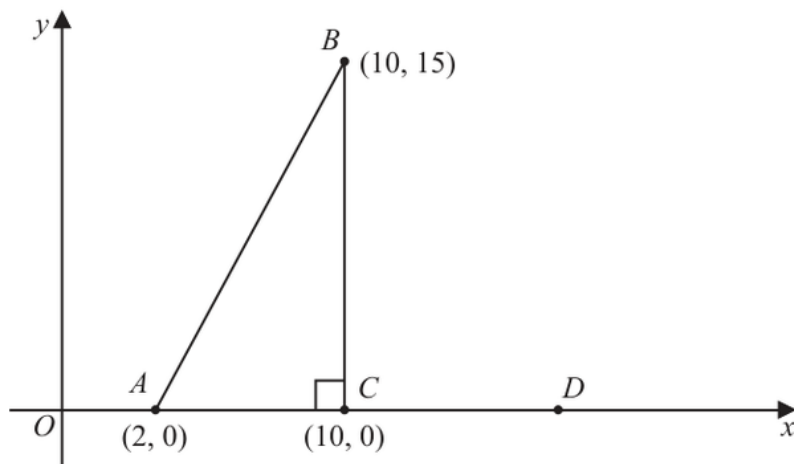


Diagram **NOT**
accurately drawn

A is the point with coordinates $(2, 0)$

B is the point with coordinates $(10, 15)$

C is the point with coordinates $(10, 0)$

D is a point on the x -axis such that $AB = AD$

Find the coordinates of D

(.....,)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$AB = \sqrt{(10 - 2)^2 + (15 - 0)^2}$$

$$= \sqrt{64 + 225} = 17$$

Since D is on the positive x -axis and $AD = 17$,

$$D = (2 + 17, 0) = (19, 0)$$

FINAL ANSWER

$(19, 0)$.

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Standard & Compound Units**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 10** A parcel in the shape of a cube is placed on a table so that the whole of one face of the parcel is in contact with the table.

The parcel has sides of length 7 cm

The pressure on the table due to the parcel is 62 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the force exerted by the parcel on the table.

..... newtons

(Total for Question 10 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 10****Marks: 3**TOPIC: **Standard & Compound Units**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

The contact area is one square face:

$$7 \times 7 = 49 \text{ cm}^2$$

$$\text{force} = \text{pressure} \times \text{area} = 62 \times 49 = 3038$$

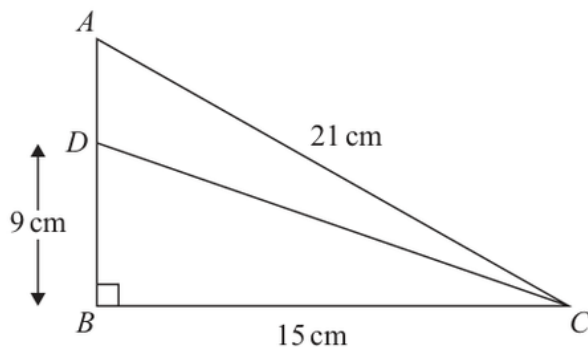
FINAL ANSWER

3038 newtons.

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

11 ABC is a right-angled triangle.Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$ Work out the size of angle ACD

Give your answer correct to one decimal place.

.....

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.**Solution**First find AB :

$$AB^2 = 21^2 - 15^2 = 216$$

$$AB = 14.696 \dots$$

$$AD = AB - DB = 14.696 \dots - 9 = 5.696 \dots$$

Angle ACB :

$$\cos ACB = \frac{15}{21}$$

$$ACB = 44.415 \dots^\circ$$

Angle DCB :

$$\tan DCB = \frac{9}{15}$$

$$DCB = 30.963 \dots^\circ$$

$$ACD = 44.415 \dots - 30.963 \dots = 13.451 \dots^\circ$$

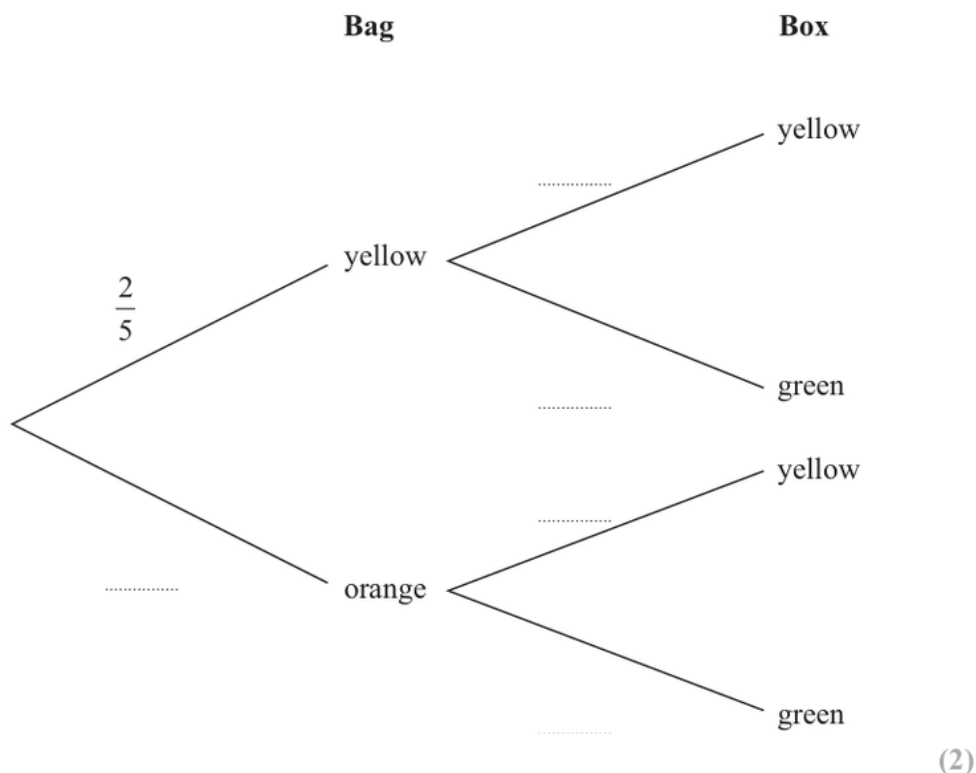
FINAL ANSWER

13.5°.

PAST PAPER QUESTION**Q#: 12** **Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 12** In a bag, there are only 2 yellow beads and 3 orange beads.
 In a box, there are only 3 yellow beads and 5 green beads.
 Chi takes at random a bead from the bag and a bead from the box.
 (a) Complete the probability tree diagram.



- (b) Work out the probability that Chi takes two beads of different colours.

.....
(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

Bag 1:

$$P(Y) = \frac{2}{5}, \quad P(O) = \frac{3}{5}$$

Bag 2:

$$P(Y) = \frac{3}{8}, \quad P(G) = \frac{5}{8}$$

The only way to get the same colour is yellow then yellow.

$$\begin{aligned} P(\text{different}) &= 1 - \frac{2}{5} \times \frac{3}{8} \\ &= 1 - \frac{3}{20} = \frac{17}{20} \end{aligned}$$

FINAL ANSWER

$$\frac{17}{20}$$

PAST PAPER QUESTION**Q#: 13****Marks: 6**TOPIC: **Expanding Brackets**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

13 (a) Expand and simplify $7x(3x + 2)(2x - 5)$
(3)(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

 $y =$
(3)

(Total for Question 13 is 6 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Expanding Brackets**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

(a)

$$\begin{aligned} & 7x(3x + 2)(2x - 5) \\ &= 7x(6x^2 - 11x - 10) \\ &= 42x^3 - 77x^2 - 70x \end{aligned}$$

(b)

$$\begin{aligned} \frac{9}{2y} + \frac{5}{7} &= 5 \\ \frac{9}{2y} &= \frac{30}{7} \\ 63 &= 60y \\ y &= \frac{21}{20} \end{aligned}$$

FINAL ANSWER

$$42x^3 - 77x^2 - 70x, y = \frac{21}{20}.$$

PAST PAPER QUESTION**Q#: 14****Marks: 2**TOPIC: **Circles, Arcs & Sectors**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 14** AOB is a sector of a circle with centre O and radius 12 cm

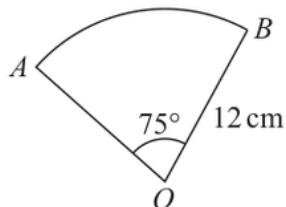


Diagram **NOT**
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 14 is 2 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$\text{area} = \frac{75}{360} \pi (12)^2$$

$$= 30\pi \approx 94.2478$$

FINAL ANSWER94.2 cm² to 3 significant figures.*Dr. Eslam Ahmed*

PAST PAPER QUESTION**Q#: 15****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

15 $\frac{16^{x-3} \times 8^{x+1}}{32^x} = 2^n$

Express n in terms of x

Give your answer in its simplest form.

 $n = \dots\dots\dots$

(Total for Question 15 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 15****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$\frac{16^{x-3} \times 8^{x+1}}{32^x} = \frac{2^{4x-12} \times 2^{3x+3}}{2^{5x}}$$
$$= 2^{4x-12+3x+3-5x} = 2^{2x-9}$$

FINAL ANSWER

$$n = 2x - 9.$$

PAST PAPER QUESTION**Q#: 16****Marks: 3**TOPIC: **Surds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 16** Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 16 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 16****Marks: 3**TOPIC: **Surds**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$\begin{aligned} & \frac{2\sqrt{7}+2}{\sqrt{7}-3} \times \frac{\sqrt{7}+3}{\sqrt{7}+3} \\ &= \frac{(2\sqrt{7}+2)(\sqrt{7}+3)}{7-9} \\ &= \frac{20+8\sqrt{7}}{-2} \\ &= -10 - 4\sqrt{7} = -10 - \sqrt{112} \end{aligned}$$

FINAL ANSWER

$$-10 - \sqrt{112}.$$

PAST PAPER QUESTION**Q#: 17****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

17 Use algebra to show that $0.9\dot{5}\dot{4} = \frac{21}{22}$

(Total for Question 17 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 17****Marks: 2**TOPIC: **Algebraic Proof**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Let

$$x = 0.95454545 \dots$$

Then

$$10x = 9.5454545 \dots$$

and

$$1000x = 954.5454545 \dots$$

Subtract:

$$990x = 945$$

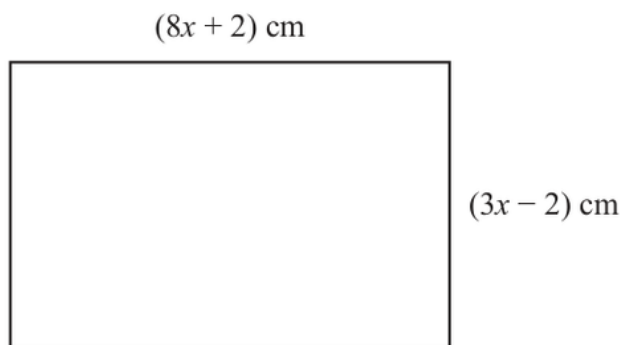
$$x = \frac{945}{990} = \frac{21}{22}$$

FINAL ANSWER

$$0.954545 \dots = \frac{21}{22}.$$

PAST PAPER QUESTION**Q#: 18****Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

18 The diagram shows a rectangle.Diagram **NOT**
accurately drawn

The rectangle has length $(8x + 2)$ cm and width $(3x - 2)$ cm
 The area of the rectangle is 500 cm^2

(a) Show that $12x^2 - 5x - 252 = 0$

(1)

(b) Hence find the value of x
 Give your answer correct to 3 significant figures.

 $x = \dots\dots\dots$

(3)

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$(8x + 2)(3x - 2) = 500$$

$$24x^2 - 10x - 4 = 500$$

$$12x^2 - 5x - 252 = 0$$

Using the quadratic formula,

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(12)(-252)}}{24}$$

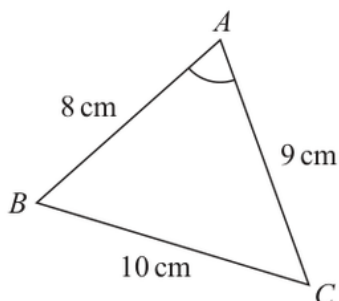
The positive solution is

$$x \approx 4.79564$$

FINAL ANSWER $x = 4.80$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 19****Marks: 3**TOPIC: **Sine, Cosine Rule & Area of Triangles**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

19 The diagram shows triangle ABC Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Sine, Cosine Rule & Area of Triangles.**Solution**Use the cosine rule for angle BAC :

$$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$$

$$\cos A = \frac{45}{144}$$

$$A = 71.790 \dots^\circ$$

FINAL ANSWER

71.8°.

PAST PAPER QUESTION

Q#: 20 Marks: 3

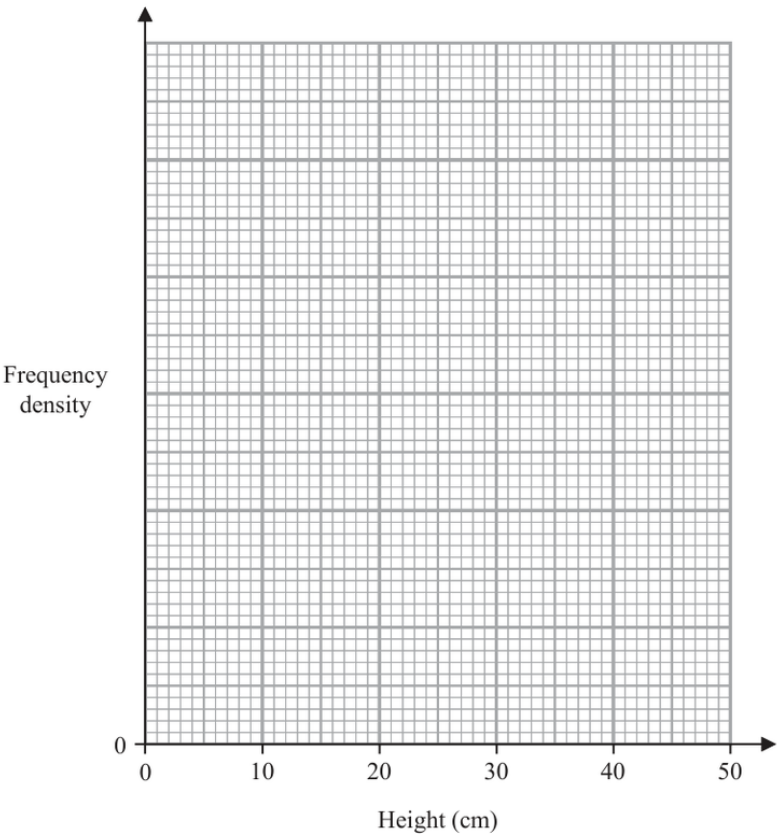
TOPIC: Histograms

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$0 < h \leq 5$	12
$5 < h \leq 15$	34
$15 < h \leq 30$	45
$30 < h \leq 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3**TOPIC: **Histograms**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**Frequency density = $\frac{\text{frequency}}{\text{class width}}$.

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$

FINAL ANSWER

draw bars with densities 2.4, 3.4, 3, 1.4.

PAST PAPER QUESTION

Q#: 20 Marks: 3

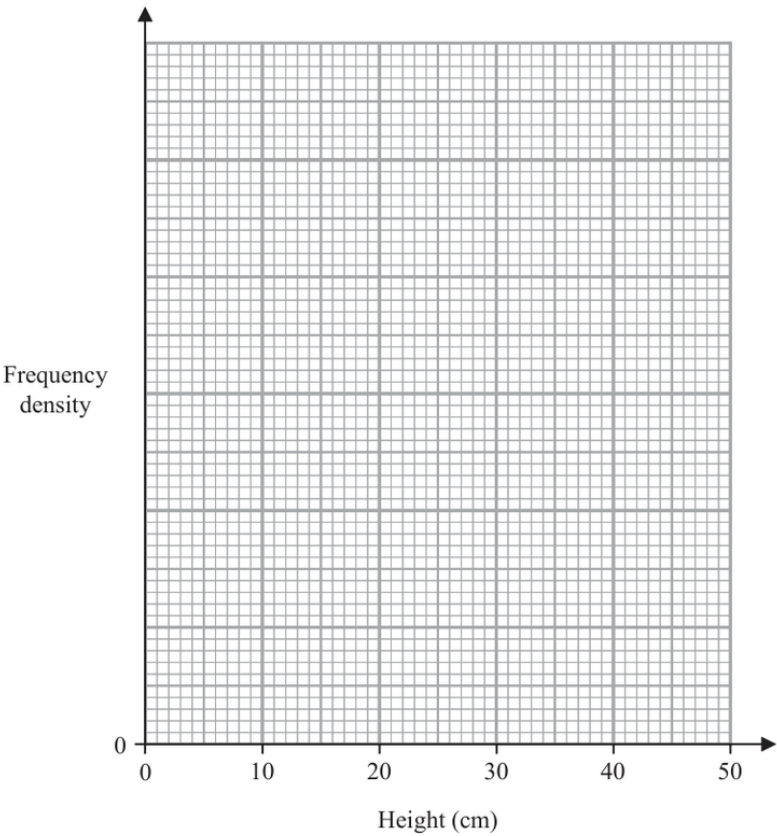
TOPIC: Histograms

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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$15 < h \leq 30$	45
$30 < h \leq 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3**TOPIC: **Histograms**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**Frequency density = $\frac{\text{frequency}}{\text{class width}}$.

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$

FINAL ANSWER

draw bars with densities 2.4, 3.4, 3, 1.4.

PAST PAPER QUESTION**Q#: 21** **Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 21** The diagram shows an equilateral triangle and a circle.

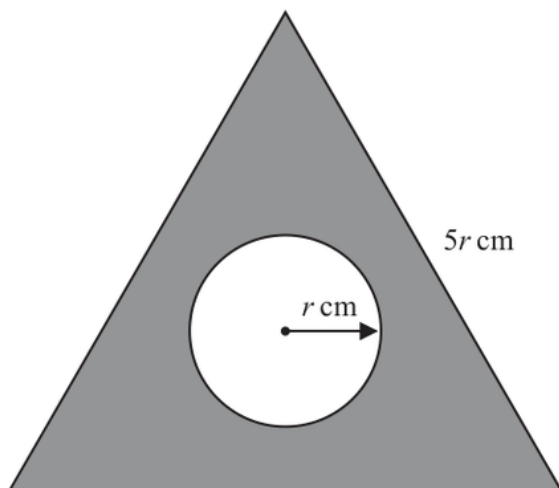


Diagram **NOT**
accurately drawn

The equilateral triangle has sides of length $5r$ cm

The circle has radius r cm

The area of the shaded region is 610π cm²

Work out the value of r

Give your answer correct to 3 significant figures.

$r = \dots\dots\dots$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

Circle area:

$$\pi r^2$$

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right)r^2 = 610\pi$$

$$r \approx 15.7926$$

FINAL ANSWER $r = 15.8$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 21** **Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

- 21** The diagram shows an equilateral triangle and a circle.

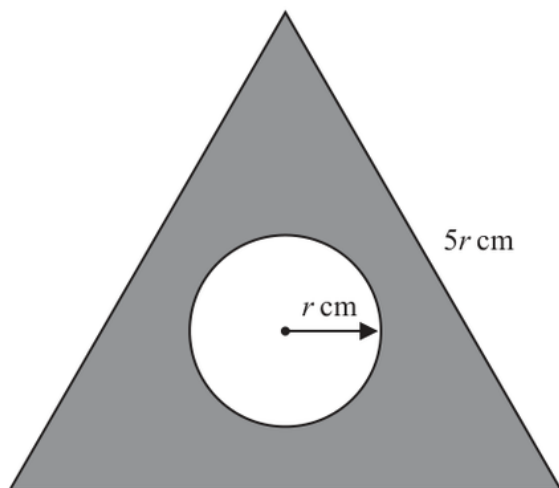


Diagram **NOT**
accurately drawn

The equilateral triangle has sides of length $5r$ cm

The circle has radius r cm

The area of the shaded region is $610\pi \text{ cm}^2$

Work out the value of r

Give your answer correct to 3 significant figures.

$r = \dots\dots\dots$

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4**TOPIC: **Area & Perimeter**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

Circle area:

$$\pi r^2$$

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right)r^2 = 610\pi$$

$$r \approx 15.7926$$

FINAL ANSWER $r = 15.8$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p , q and r are integers.

Find the value of q

$q = \dots\dots\dots$

(Total for Question 22 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

FINAL ANSWER

$$q = 4.$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p , q and r are integers.

Find the value of q

$q = \dots\dots\dots$

(Total for Question 22 is 2 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 22****Marks: 2****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

FINAL ANSWER

$$q = 4.$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

23 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

There are 2 pink buttons.

At least one pink button is still in the box unless both pink buttons are taken.

So use the complement.

The probability that both pink buttons are taken is

$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

FINAL ANSWER

$$\frac{34}{35}$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

23 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

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At least one pink button is still in the box unless both pink buttons are taken.

So use the complement.

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$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

FINAL ANSWER

$$\frac{34}{35}$$

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 24 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x - 2)^2 - 4) + 23$$

$$= 5(x - 2)^2 + 3$$

FINAL ANSWER

$$5(x - 2)^2 + 3.$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 24 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Completing the Square**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x - 2)^2 - 4) + 23$$

$$= 5(x - 2)^2 + 3$$

FINAL ANSWER

$$5(x - 2)^2 + 3.$$

PAST PAPER QUESTION**Q#: 25****Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

25 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 25 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 25****Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected from Ratio Toolkit to Algebraic Fractions.**Method**

$$a = \frac{2x + 5}{1 - x}, \quad x = \frac{5 - 2y}{3y}$$

Substitute for x .

Numerator:

$$\begin{aligned} 2x + 5 &= 2 \left(\frac{5 - 2y}{3y} \right) + 5 \\ &= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y} \end{aligned}$$

Denominator:

$$\begin{aligned} 1 - x &= 1 - \frac{5 - 2y}{3y} \\ &= \frac{3y - 5 + 2y}{3y} = \frac{5y - 5}{3y} = \frac{5(y - 1)}{3y} \end{aligned}$$

Therefore

$$a = \frac{\frac{10 + 11y}{3y}}{\frac{5(y - 1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

FINAL ANSWER

$$\frac{10 + 11y}{5(y - 1)}$$

PAST PAPER QUESTION**Q#: 25** **Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

$$25 \quad a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

 $a = \dots\dots\dots$

(Total for Question 25 is 3 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 25****Marks: 3**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected from Ratio Toolkit to Algebraic Fractions.**Method**

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Therefore

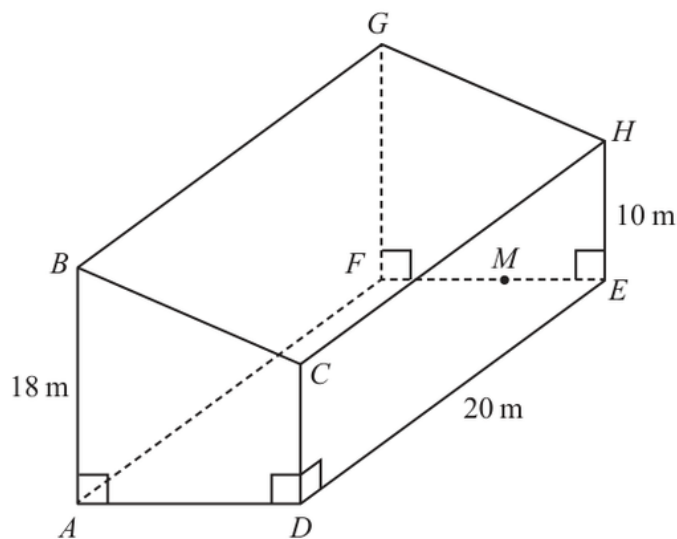
$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

FINAL ANSWER

$$\frac{10 + 11y}{5(y - 1)}$$

PAST PAPER QUESTION**Q#: 26** **Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

26 The diagram shows prism $ABCDEFGH$ Diagram **NOT**
accurately drawn

$$AB = FG = 18\text{ m} \quad DC = EH = 10\text{ m} \quad AD = FE$$

$$DE = CH = AF = BG = 20\text{ m}$$

$$\text{angle } BAD = \text{angle } ADC = \text{angle } CDE = 90^\circ$$

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that $\text{angle } GMF = 60^\circ$

calculate the angle of elevation of G from D

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**From triangle GMF ,

$$\tan 60^\circ = \frac{FG}{FM} = \frac{18}{FM}$$

$$FM = \frac{18}{\tan 60^\circ} = 6\sqrt{3}$$

Since $FM = \frac{3}{5}FE$,

$$FE = 10\sqrt{3}$$

The horizontal distance from D to the point below G is

$$\sqrt{20^2 + (10\sqrt{3})^2} = 10\sqrt{7}$$

If θ is the angle of elevation,

$$\tan \theta = \frac{18}{10\sqrt{7}}$$

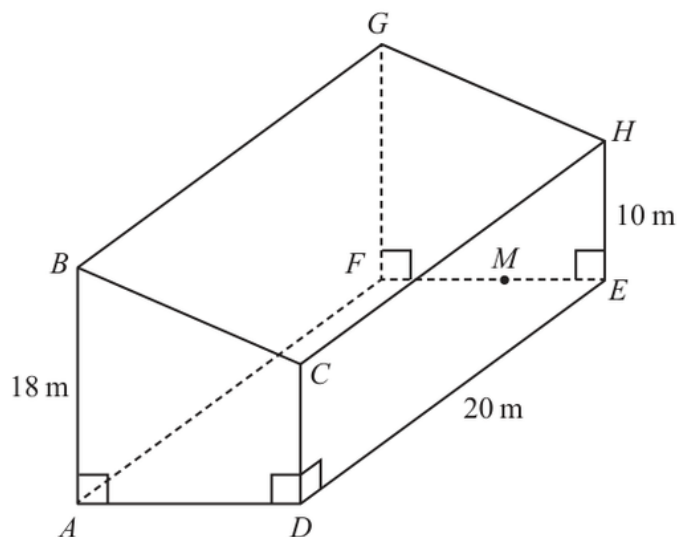
$$\theta \approx 34.2289^\circ$$

FINAL ANSWER

34.2°.

PAST PAPER QUESTION**Q#: 26** **Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

26 The diagram shows prism $ABCDEFGH$ Diagram **NOT**
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Given that $\text{angle } GMF = 60^\circ$

calculate the angle of elevation of G from D

Give your answer correct to one decimal place.

Show your working clearly.

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

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If θ is the angle of elevation,

$$\tan \theta = \frac{18}{10\sqrt{7}}$$

$$\theta \approx 34.2289^\circ$$

FINAL ANSWER

34.2°.

PAST PAPER QUESTION**Q#: 27** **Marks: 5**TOPIC: **Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

Show your working clearly.

(Total for Question 27 is 5 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 27****Marks: 5****TOPIC: Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Diagonals of a rhombus bisect each other at right angles.

Midpoint of AC :

$$\left(\frac{16 + 24}{2}, \frac{-9 + 15}{2} \right) = (20, 3)$$

Gradient of AC :

$$\frac{15 - (-9)}{24 - 16} = 3$$

So gradient of BD is $-\frac{1}{3}$.

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

FINAL ANSWER

$$x + 3y - 29 = 0.$$

PAST PAPER QUESTION**Q#: 27** **Marks: 5**TOPIC: **Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD

Give your answer in the form $ax + by + c = 0$ where a , b and c are integers.

Show your working clearly.

(Total for Question 27 is 5 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 27****Marks: 5****TOPIC: Coordinate Geometry**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

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Midpoint of AC :

$$\left(\frac{16 + 24}{2}, \frac{-9 + 15}{2} \right) = (20, 3)$$

Gradient of AC :

$$\frac{15 - (-9)}{24 - 16} = 3$$

So gradient of BD is $-\frac{1}{3}$.

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

FINAL ANSWER

$$x + 3y - 29 = 0.$$

PAST PAPER QUESTION**Q#: 28****Marks: 4**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

28 Solve $\frac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \frac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(Total for Question 28 is 4 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 28****Marks: 4**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

$$2x(5x-4) = 0$$

The original fractions do not allow $x = 0$, so

$$x = \frac{4}{5}$$

FINAL ANSWER

$$x = \frac{4}{5}.$$

PAST PAPER QUESTION**Q#: 28****Marks: 4**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

28 Solve $\frac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \frac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

 $x = \dots\dots\dots$

(Total for Question 28 is 4 marks)

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 28****Marks: 4**TOPIC: **Algebraic Fractions**

Nov 2025 4WM1H - Nov2025 | 4WM1H | Unit 1

WORKED SOLUTION*Dr. Eslam Ahmed***Solution**

Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

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The original fractions do not allow $x = 0$, so

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FINAL ANSWER

$$x = \frac{4}{5}.$$